

A Model Predictive Control Scheme for a Single-Phase Integrated Battery Charger With Active Power Decoupling for EV Application

Deeksha Bhule^{1b}, *Student Member, IEEE*, and R. Sudharshan Kaarthik^{1b}, *Senior Member, IEEE*

Abstract—Integrated battery chargers (IBCs) are one of the best alternatives among the numerous types of onboard electric vehicle (EV) chargers available, considering the size and weight constraints of EVs. In this article, a model predictive control (MPC) scheme for a single-phase IBC is proposed for the first time. The IBC repurposes the motor drive, which consists of a three-phase induction motor and a three-phase inverter, for charging purposes. The proposed MPC scheme is implemented with the objective of minimization of switching frequency using double prediction algorithm, along with maintaining a sinusoidal grid current profile at unity power factor (UPF) during charging mode. Furthermore, the proposed scheme addresses the issue of ripple power at the dc link with twice the grid frequency, which is inherently present in a single-phase IBCs. This ripple component is effectively suppressed by using active power decoupling by reutilizing one of the stator windings of the induction motor along with the inverter legs without any additional hardware requirement. Additionally, the power reversal capability of IBC is also demonstrated. The control scheme was implemented on hardware by using Texas instruments' DSP TMS320F28335.

Index Terms—Active power decoupling, electric vehicles (EVs), integrated battery chargers (IBCs), model predictive control (MPC).

I. INTRODUCTION

IN RECENT years, there has been a rapid increase in the popularity of electric vehicles (EVs) due to a number of factors, including concerns over air pollution and climate change, advancements in battery technology, and government incentives and regulations [1]. Reducing greenhouse gas emissions and combating climate change are some of the primary drivers to shift to EVs. The transportation sector is a major contributor to global carbon emissions, and EVs offer a way to significantly reduce these emissions by using electricity generated from renewable sources.

The commercialization of EVs faces challenges such as limited driving range, high costs, weight, lack of charging

Manuscript received 25 August 2023; revised 7 December 2023; accepted 28 December 2023. Date of publication 8 January 2024; date of current version 16 February 2024. This work was supported by the Kerala State Council for Science, Technology and Environment (KSCSTE) through the project "Novel high power integrated battery charging for electric cars and e-bikes." Recommended for publication by Associate Editor M. Saeedifard. (*Corresponding author: R. Sudharshan Kaarthik.*)

The authors are with the Department of Avionics, Indian Institute of Space Science and Technology, Thiruvananthapuram, Kerala 695547, India (e-mail: deekshabhule.21@res.iist.ac.in; sudharshan.kaarthik@iist.ac.in).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TPEL.2024.3350991>.

Digital Object Identifier 10.1109/TPEL.2024.3350991

TABLE I
COMPARISON OF VARIOUS TOPOLOGIES OF INTEGRATED BATTERY CHARGERS [3]

Ref.	No. of machine phases	Type of supply	No. of converter legs	Pulsating torque	Hardware reconfiguration needed
[4]	3	1-ph	3	Yes	Yes
[5]	9	1-ph/3-ph	9	No	No
[6]	6	1-ph/3-ph	6	Yes	Yes
[7]	6	1-ph/3-ph	6	No	Yes
[8]	5	1-ph/3-ph	5	Yes	Yes
[9]	6	1-ph/3-ph	8	Yes	Yes
[10]	6	3-ph	6	No	No
[11]	9	3-ph	6	No	Yes
[12]	6	3-ph	3	No	Yes
[13]	3	1-ph/3-ph	2x3	No	Yes
[14]	3	DC	4	No	No
[15]	3	3-ph	3 VSI+3CSI	No	No
[16], [17], [18], [19]	3	3-ph	3	No	Yes

infrastructure, and long charging times. The use of an on-board integrated battery charger offers several benefits, including space savings, weight reduction, increased efficiency, and cost savings. It eliminates the need for a separate charging unit, making EVs more affordable and suitable for daily use. For the vast majority of integrated battery chargers, finding an appropriate strategy in terms of resilience and zero torque production during charging is exceedingly difficult. Primarily, induction motors, switched-reluctance motors, and permanent magnet motors are used in integrated battery charger (IBC) schemes [2]. IBC can be classified either as a single-phase IBC, or a three-phase IBC, depending on the grid supply utilized for charging the battery. An essential condition for grid-connected systems is to function with a unity input power factor. However, in a single-phase sinusoidal grid-connected energy conversion systems, such as the integrated charger configuration used in this article, a low-frequency power ripple is inherently present in the dc voltage bus. These single-phase charging systems, exhibiting twice the grid frequency power ripple on the dc side, cause an increase in the battery temperature and, consequently, a reduction in the life span of the battery.

Several on-board integrated battery chargers have been proposed in the literature [20], [21], [22], [23], [24], [25], [26], [27]. The comparisons of various existing topologies of integrated battery chargers in the literature is shown in Table I. In [11], the three-phase inverter used for controlling the motor during driving was reutilized for the purpose of charging the battery. However, there was a need for an external inductor for interfacing with the grid to charge the battery, which in turn

may lead to an increase in the weight and cost of the vehicle. The need for an external inductor as a grid interface inductor for charging purpose was eliminated in [28] as the two stator windings (connected in series) of the propulsion motor were used. However, the issue of ripple on the dc side was not explored in [28]. This issue was later explored in [16] using the third stator winding of the motor for active power decoupling. However, only simulation studies have been provided in it. In [17] and [18], for a similar configuration presented in [16], the two stator windings were connected in parallel instead of series to be used as grid interfacing inductance. In addition to the hardware architecture of the EV, an efficient control method that achieves the desired goals is essential.

The work proposed in this article focuses on model predictive control of a single-phase integrated battery charger, which utilizes the stator windings of the motor and an existing three-phase VSI for battery charging [19]. In this configuration, in order to maximize the charging capability, the two stator windings are connected in parallel, and hence, double the rated current of the motor can be drawn from the grid at UPF for charging the battery. This parallel combination of the windings is used as the grid-interface inductance. The issue of eliminating the 100 Hz harmonic ripple on the dc side is tackled by using the ripple-power cancellation method using an energy-storage inductor. For this purpose, the third winding of the motor is used as the energy storage inductor for active power decoupling, hence, eliminating the need for an external component. The current through it is controlled by using the MPC scheme. The proposed single-phase IBC control scheme also ensures zero torque production during battery charging.

The novel contributions in this article can be summarized as follows.

- 1) An MPC scheme for controlling the grid current and the decoupling current (current through the third motor winding) of a single-phase IBC system is proposed for the first time in this article.
- 2) The uniqueness of the proposed method is that the optimal switching state is chosen to minimize the switching frequency, in other words, the dwell time of the most optimal switching state is also predicted (double prediction) in the proposed scheme.
- 3) The proposed control approach allows the inverter to operate at a lower switching frequency than that of the sinusoidal pulsewidth modulation (SPWM) technique for a given current ripple while maintaining the requisite dc-link voltage for battery charging.
- 4) Reduced switching losses are a benefit of operating the inverter at a lower switching frequency.
- 5) With the proposed MPC technique, the current through the grid-interface stator winding is controlled to be sinusoidal and in phase with the grid voltage (UPF) during charging.
- 6) Additionally, the current via the third stator winding is effectively controlled to follow the reference to eliminate the 100 Hz harmonic ripple flowing into the battery by using active power decoupling.
- 7) Furthermore, the vehicle to grid operation (V2G), i.e., power reversal capability is ensured with the proposed scheme.

The rest of this article is organized as follows. Section II discusses the single-phase IBC configuration, zero torque production, and the modes of operation. Section III describes the active-power decoupling technique using c-phase winding. Section IV describes the proposed MPC control scheme. The experimental and simulation results validating the proposed scheme is given in Section V. Finally, Section VI concludes this article.

II. SINGLE-PHASE INTEGRATED BATTERY CHARGER

Fig. 1 shows the schematic of a single-phase (IBC) [18]. The IBC repurposes the motor drive components comprising of a three-phase induction motor and a three-phase two-level voltage source inverter (VSI) for battery charging applications. In this system, the available VSI is used as a unity power factor front-end converter (grid-connected ac-dc converter). The machine windings are connected to the utility grid, and the leakage inductances of the machine windings are used as the grid interface inductance. For this purpose, the two stator windings of the three-phase induction motor, a and b, are connected in parallel between the grid and the inverter. The use of machine stator windings as interface inductance eliminates the requirement of additional bulky low-frequency inductors. In this system, the leakage inductance of each motor winding is assumed to be identical. Hence, it can be said that the balanced current drawn from the grid at UPF is divided equally amongst both the stator windings (a and b) connected in parallel. The resultant mmf produced by the current flowing through the windings a and b is pulsating in nature and does not produce a rotating magnetic field. Hence, the torque produced by the motor during the charging mode is zero. The different modes of operation in the IBC are explained in the following section.

A. Modes of Operation

The single-phase IBC operates in two modes as follows: 1) the charging mode and 2) the driving mode. The circuit uses two single pole double throw (SPDT) contactors. The connections of SPDT contactors vary depending on the mode in which they are operated. The driving mode and the charging mode are discussed in the section that follows.

1) *Driving Mode:* During the driving mode, c_2 is connected to the terminal-2' (S_3), and b_1 is connected to the terminal-1' (S_2). With this, c_2 , b_2 , and a_2 form the star connection, and the other ends of the winding, c_1 , b_1 , and a_1 , are connected to the C, B, and A leg of the inverter, respectively. The motor is operated in star in this mode of operation. Therefore, the system operates as a conventional three-phase VSI-fed motor connected in star, as shown in the Fig. 1.

2) *Charging Mode:* In the charging mode, the vehicle is connected to the grid via the contactor S_1 , as depicted in Fig. 1, c_2 is connected to terminal-2 (S_3), and b_1 is connected to terminal-1 (S_2) to enable charging mode. This results in a parallel connection of a and b windings between the grid and the inverter. The parallel combination of stator windings, a-phase, and b-phase functions as the interface inductance between the grid and the inverter. The third motor winding (c-phase) is

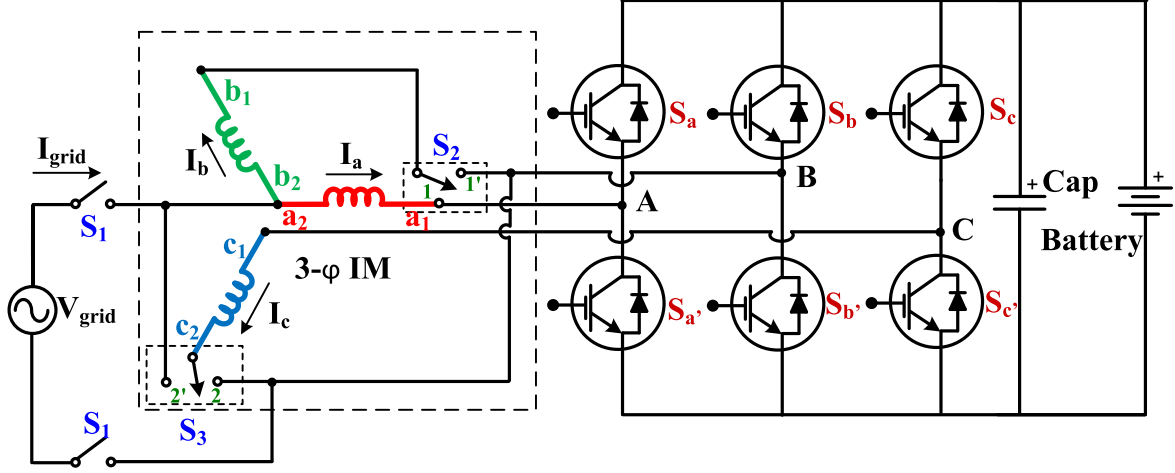


Fig. 1. Schematic of a single-phase integrated battery charger.

connected between the B and C legs of the inverter. By controlling the switching state of inverter leg-C, the current through the third stator winding (c) of the motor is controlled, and the 100 Hz power ripple flowing into the battery is eliminated. In the following section, the concept of active-power decoupling using the c-phase winding is explained.

III. ACTIVE POWER DECOUPLING

A. Need for Active Power Decoupling

The fundamental requirement of grid-connected systems is maintaining the power factor unity during operation. However, the presence of inherent power ripple in the dc voltage bus poses a problem for single-phase sinusoidal grid-connected systems. Assume that the grid voltage, V_{grid} and the current drawn by the grid at unity power factor, I_{grid} can be written as

$$V_{\text{grid}} = V_m \cos(\omega t) \quad (1)$$

$$I_{\text{grid}} = I_m \cos(\omega t) \quad (2)$$

where V_m is the peak value of the grid voltage and I_m is the peak value of the current drawn from the grid. The instantaneous power drawn from the grid, P_{grid} can be written as

$$P_{\text{inst}} = V_{\text{grid}} \times I_{\text{grid}} \quad (3)$$

$$= V_m \cos(\omega t) \times I_m \cos(\omega t) \quad (4)$$

$$= \frac{V_m I_m}{2} + \frac{V_m I_m}{2} \cos(2\omega t). \quad (5)$$

In (5), the first term represents the average power that flows into the battery, and the second term represents the oscillating power. Hence, it can be said that an oscillating power of frequency 2ω (100 Hz) flows from the grid during UPF operation to the dc-link in case of single-phase front-end converter.

B. Elimination of 100 Hz Power Component Flow to DC-Link

In a single-phase grid-connected battery charging system, it is necessary to reduce the 100 Hz ripple power flowing into the

battery during charging. By minimizing the amount of ripple, the risk of damage to the battery can be reduced, ensuring that the battery can operate safely and efficiently over its lifetime. Several strategies have been proposed to eliminate this 100 Hz ripple in the dc-link. Using an additional inductance for decoupling [29] is one of the simplest methods described in the literature for removing this 100 Hz ripple, compared to those involving the addition of switching circuits [30]. But the scheme in [29] suggests the use of an extra inductor for active power decoupling, which in turn can cause an increase in the weight and cost of the system. In the active power decoupling technique used in this article, the ripple power flowing from the grid to the dc-link is bypassed through the third stator winding (c) of the motor, and the power going into the load is made approximately constant. No external circuits are needed in the system since the stator winding is utilized for active-power decoupling.

The power balance equation in the system can be written as

$$P_{\text{inst}} = P_{\text{load}} + P_{\text{grid}} + P_c \quad (6)$$

where, P_{inst} is the instantaneous power from the grid, P_{load} is the power flowing to the load, P_{grid} is the power corresponding to the energy stored in the grid interface inductance, P_c is the power corresponding to the energy stored in the c-phase winding.

Assume that the current of the energy-storage inductor (c-phase winding) takes the form of

$$I_c = I_{cm} \cos(\omega t + \alpha) \quad (7)$$

where I_{cm} is the peak value of the current through c-phase winding, α is the angle of the c-phase winding current with respect to the grid.

The energy stored by the c-phase winding can be expressed as

$$E_c = \frac{1}{2} L_c I_c^2(t) \quad (8)$$

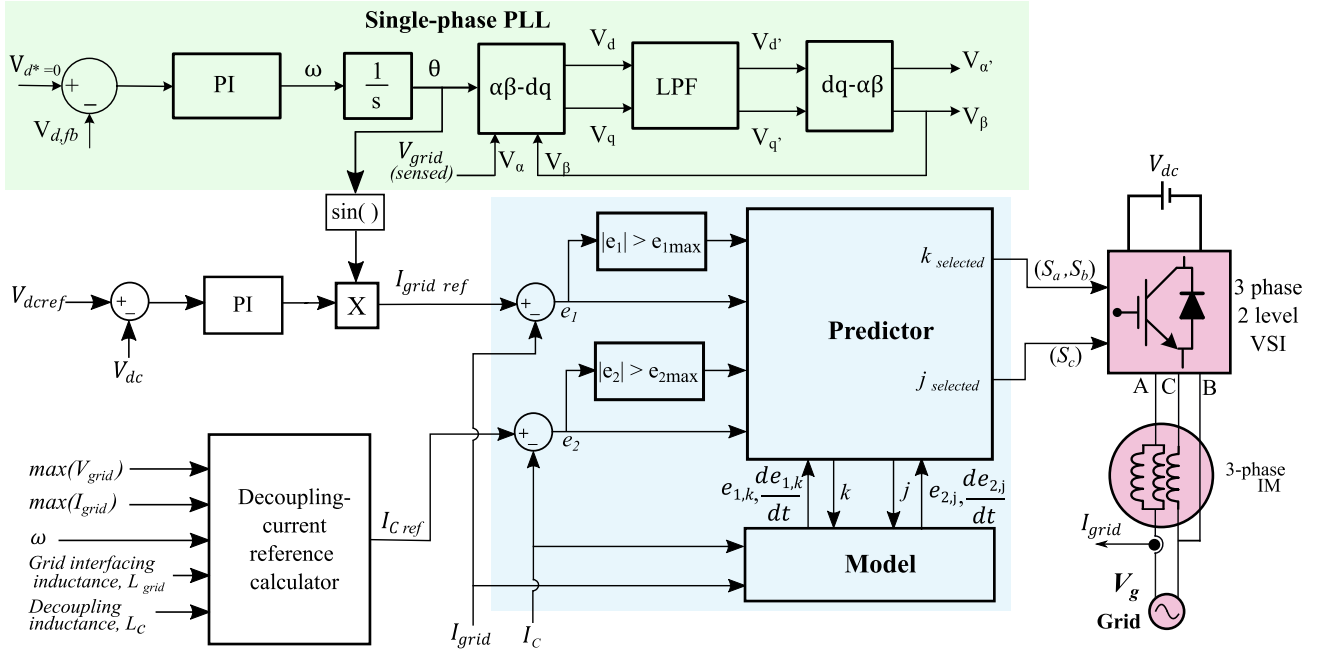


Fig. 2. Block diagram of the proposed MPC control scheme for Single-phase IBC.

where L_c is the inductance of c-phase winding. The corresponding power can be written as

$$P_c = -\frac{1}{2}\omega L_c I_{cm}^2 \sin(2\omega t + 2\alpha). \quad (9)$$

Similarly, the energy stored by the grid interface inductance, L_{grid} can be expressed as

$$E_{grid} = \frac{1}{2}L_{grid} I_{grid}^2(t). \quad (10)$$

The corresponding power can be written as

$$P_{grid} = -\frac{1}{2}\omega L_{grid} I_m^2 \sin(2\omega t). \quad (11)$$

Substituting the above derived equations in power balance equation we get

$$\begin{aligned} \frac{V_m I_m}{2} + \frac{V_m I_m}{2} \cos(2\omega t) &= P_{load} - \frac{1}{2}\omega L_{grid} I_m^2 \sin(2\omega t) \\ &- \frac{1}{2}\omega L_c I_{cm}^2 \sin(2\omega t + 2\alpha). \end{aligned} \quad (12)$$

The load power, P_{load} is assumed to be constant. Hence, to negate the ripple power, the equivalent power in the energy-storage inductor (L_c) must be equal to the sum of the oscillating power from the grid and power from grid-interface inductance. Hence, the above equation becomes

$$\begin{aligned} \frac{V_m I_m}{2} \cos(2\omega t) + \frac{1}{2}\omega L_{grid} I_m^2 \sin(2\omega t) &= \\ \frac{1}{2}\omega L_c I_{cm}^2 \sin(2\omega t + 2\alpha). \end{aligned} \quad (13)$$

Simplifying above equation I_{cm} and α can be calculated as shown below

$$I_{cm} = \sqrt{\frac{\sqrt{(I_m V_m)^2 + (\omega L_{grid} I_m^2)^2}}{\omega L_c}} \quad (14)$$

$$\alpha = -\frac{\beta}{2} - \frac{\pi}{4} \quad (15)$$

where $\beta = \tan^{-1}\left(\frac{\omega L_{grid} I_m}{V_m}\right)$.

Substituting the value of I_c and β in equation, the required current for the c-phase winding to decouple the ripple power is attained. This current required for decoupling is controlled by using the model predictive control scheme is discussed in the next section.

IV. PROPOSED MODEL PREDICTIVE CONTROL SCHEME

The model predictive control for a power electronic converter is based on the finite control set (FCS). This is due to the fact that the state of an IGBT/MOSFET switch in a power electronic converter is constrained to either 0 or 1. The control set is formed on the basis of the number of switches that can operate independently. The fundamental idea underlying FCS-MPC is that the controlled variable is predicted for all the possible switching states of the converter, and the switching state that satisfies the desired objective function is chosen. The block diagram of the proposed MPC scheme for IBC is shown in Fig. 2. A voltage controller is required to maintain the voltage across the dc-link. In the proposed system, the grid current reference is generated using the voltage controller. In the practical case of an EV charger, the battery current reference provided by the battery management system (BMS), will be controlled using the proposed model predictive scheme. In the proposed model

predictive control scheme, the grid current, and the decoupling current are controlled using double prediction. The current loop is triggered only when the error between the reference, and actual current crosses the specified value. After the activation of the model predictive control algorithm, it gives the required switching pulses for the three-phase inverter to minimize the switching frequency. The proposed model predictive current controller is discussed in detail in the second section.

A. Phase-Locked Loop (PLL)

The purpose of the single-phase phase locked loop (PLL) in the proposed IBC system is to synchronize the angle (θ) with the grid voltage, V_{grid} that has a fixed frequency and phase relationship. The block diagram for the single-phase PLL is shown in Fig. 2. The grid voltage is sensed using a voltage sensor and assumed to be equal to V_α . The angle obtained by PLL, θ , is used to form the reference currents for grid current and decoupling current through the c-phase winding. These reference currents generated are then given to the MPC block. The model predictive current controller is used to control two currents, the current drawn from the grid, I_{grid} , and the current through the c-phase winding, I_c for active power decoupling, as discussed in the next section.

B. Model Predictive Current Controller (MPCC)

The basic idea behind MPC is to use a mathematical model of the system being controlled to predict the future behavior of the system based on its current state and input signals. The controller then uses this prediction to calculate an optimal control action to meet the cost function. This cost function can be designed to take into account various performance objectives, such as minimizing switching frequency, minimizing current ripple, reducing harmonic distortion, or maximizing efficiency. This work focuses on minimizing the switching frequency of the inverter switches while controlling the currents to follow the reference waveform [31]. Hence, the objective of the cost function of the proposed MPC scheme is to maximize the switching period (Δt) while considering the number of commutations required to obtain the next switching state.

In the conventional SPWM scheme, the voltage vectors are applied in the fixed sequence. In contrast, the proposed model predictive scheme introduces a dynamic approach wherein the sequence of the applied voltage vectors in a switching cycle is not fixed. The voltage vector, which provides the minimum switching frequency, is selected.

Unlike the traditional MPC schemes, the proposed scheme works on the double-prediction algorithm [31]. The model predictive current loop gets triggered only when the error between the reference and the actual currents crosses the specified value (current error boundary). In this algorithm, at any switching instant, immediately after a new switching state is activated, the future values of the reference and actual current are predicted. Using these values of predicted currents, the next optimum switching state is computed, along with the prediction of the time instant at which the error will cross the specified current error boundary. This prediction of the time instant and the

computation of the switching state before the occurrence of the event helps in avoiding the response delay, and the current overshoot that may occur due to delay in the computation. In this article, the control variables for MPCC are the grid current (I_{grid}) and the decoupling current (I_c). The inverter switches of leg-A and leg-B are switched to control the grid current (I_{grid}), while the switches of leg-C of the inverter are modulated to control the decoupling current (I_c). As the motor stator windings a and b, are connected in parallel, the voltage across the two windings will be the same. The generalized equation for stator voltages for an induction motor can be written as [32]

$$\begin{aligned} V_{su} = & R_s I_{su} + (L_s + L_m) \frac{dI_{su}}{dt} + L_m \cos\left(\frac{2\pi}{3}\right) \frac{dI_{sv}}{dt} \\ & + L_m \cos\left(\frac{4\pi}{3}\right) \frac{dI_{sw}}{dt} + M \frac{d}{dt} I_{ru} \cos \theta_r + M \frac{d}{dt} I_{rv} \\ & \cos\left(\theta_r + \frac{2\pi}{3}\right) + M \frac{d}{dt} I_{rw} \cos\left(\theta_r + \frac{4\pi}{3}\right) \end{aligned} \quad (16)$$

where u represents any of the three phases - a, b, and c. v and w represent the corresponding phases lagging u by 120° and 240° , respectively. V_{su} represents the voltage across the stator winding of phase u, and I_{su} represents the currents through the stator winding of phase u. I_{ru} is the current through the rotor windings of phase u. L_m is the mutual inductance between the stator windings, L_s is the stator leakage inductance of each winding, and M is the mutual inductance between the rotor and stator windings. θ_r is the angle between the rotor and stator flux.

It is assumed that the stator resistances and the leakage inductances are equal in all the stator windings. For the single-phase IBC system presented in this article, the voltage across the stator winding and the current in the winding, a and b, are equal, i.e., $V_{sa} = V_{sb}$, $I_{sa} = I_{sb}$. Substituting these equations in (5) and (7), the voltage across stator windings a and c can be represented as shown below

$$\begin{bmatrix} V_{sa} \\ V_{sc} \end{bmatrix} = \begin{bmatrix} R_s + p(L_s + 0.5L_m) & -0.5L_m p \\ -L_m p & R_s + p(L_s + L_m) \end{bmatrix} \begin{bmatrix} I_{sa} \\ I_{sc} \end{bmatrix}. \quad (17)$$

The grid current, I_{grid} is the sum of the currents flowing in the stator windings, a and b. The derivative of the predicted grid current at any possible switching state, k, can be written as

$$\frac{dI_{\text{grid}}(t, k)}{dt} = \frac{V_{\text{grid}} - V_{AB, k}}{L_{\text{grid}}} \quad (18)$$

where k is the possible switching states for legs A and B of the inverter, V_{grid} is grid voltage, and $V_{AB, k}$ is the voltage between leg A and B of the inverter at any switching state, k.

The derivative of the reference grid current can be written as

$$\frac{dI_{\text{gref}}}{dt} = \frac{I_{\text{gref}}(n+1) - I_{\text{gref}}(n)}{\Delta t_1}. \quad (19)$$

The derivative of the predicted error of grid current can be written as

$$\frac{de_1(t, k)}{dt} = \frac{dI_{\text{grid}}(t, k)}{dt} - \frac{dI_{\text{gref}}}{dt}. \quad (20)$$

At any given time, t , the error in the grid current at a switching state, k , can be written as

$$e_1(t, k) = e_1(t_0) + \frac{de_1(t, k)}{dt} \Delta t_1 \quad (21)$$

where $e_1(t_0)$ is the error in the predicted grid current at time, t_0 . By squaring the error function, we get

$$(e_1(t, k))^2 = (e_1(t_0))^2 + 2e_1(t_0) \frac{de_1(t, k)}{dt} \Delta t_1 + \left(\frac{de_1(t, k)}{dt} \Delta t_1 \right)^2 \quad (22)$$

for $e_1(t, k) = e_1(t_0)$, the above equation can be rewritten as

$$2e_1(t_0) \frac{de_1(t, k)}{dt} \Delta t_1 + \left(\frac{de_1(t, k)}{dt} \Delta t_1 \right)^2 = 0. \quad (23)$$

Solving the above equation for Δt_1 , we get the objective function to be maximized as

$$\Delta t_1 = \frac{-p_1}{p_2} \frac{1}{n_c} \quad (24)$$

where n_c is the number of commutations, $p_1 = 2e_1(t_0) \frac{de_1(t, k)}{dt}$ and $p_2 = \left(\frac{de_1(t, k)}{dt} \right)^2$.

By using (26), the switching state, k_{selected} corresponding to the maximum objective function, Δt_1 is obtained. The switching state, k_{selected} (for the switching of legs A and B of the inverter) is used to determine the switching state of the C leg of the inverter at any time. The derivative of the predicted c-winding current can be written as

$$\frac{dI_c(t, j)}{dt} = \frac{V_C(t, j) - V_B(t, k_{\text{selected}})}{L_c} \quad (25)$$

where j is the possible switching state of leg C of the inverter, $V_C(t, j)$ is the pole voltage at C leg of the inverter at any possible switching state, j , and $V_B(t, k_{\text{selected}})$ is the pole voltage at B leg of the inverter corresponding to k_{selected} to maximize Δt_1 .

The derivative of the predicted current error of decoupling current can be written as

$$\frac{de_2(t, j)}{dt} = \frac{dI_c(t, j)}{dt} - \frac{dI_{\text{cref}}}{dt}. \quad (26)$$

The derivative of the reference decoupling current can be written as

$$\frac{dI_{\text{cref}}}{dt} = \frac{I_{\text{cref}}(n+1) - I_{\text{cref}}(n)}{\Delta t_2}. \quad (27)$$

Similarly, at any instant t , the error in the c winding current can be written as

$$e_2(t, j) = e_2(t_0) + \frac{de_2(t, j)}{dt} \Delta t_2 \quad (28)$$

where $e_2(t_0)$ is the value of the current error in the c-winding current at time, t_0 .

By squaring the error function from the above equation, we get

$$(e_2(t, j))^2 = (e_2(t_0))^2 + 2e_2(t_0) \frac{de_2(t, j)}{dt} \Delta t_2$$

TABLE II

SYSTEM PARAMETERS USED FOR SIMULATION AND EXPERIMENTAL STUDIES

Parameter	Value
Input voltage	220 V rms
DC link voltage	400 V
DC link capacitor	2.2 mF
Load resistance	100 Ω
Motor stator leakage inductance	14 mH
Motor rotor leakage inductance	14 mH
Motor mutual inductance	80 mH
Stator resistance	1.05 Ω
Rotor resistance	1.97 Ω

$$+ \left(\frac{de_2(t, j)}{dt} \Delta t_2 \right)^2. \quad (29)$$

For $e_2(t, j) = e_2(t_0)$, the above equation can be rewritten as

$$2e_2(t_0) \frac{de_2(t, j)}{dt} \Delta t_2 + \left(\frac{de_2(t, j)}{dt} \Delta t_2 \right)^2 = 0. \quad (30)$$

Simplifying the above equation for Δt_2 , the cost function to be maximized for switching of the C leg of the inverter can be written as

$$\Delta t_2 = \frac{-q_1}{q_2} \frac{1}{n_c} \quad (31)$$

where n_c is the number of commutations, $q_1 = 2e_2(t_0) \frac{de_2(t, j)}{dt}$ and $q_2 = \left(\frac{de_2(t, j)}{dt} \right)^2$.

Thus, the two switching states, k_{selected} (switching state selected for the switching of inverter legs, A and B) and j_{selected} (switching state selected for the switching of inverter leg, C) are the required switching states. Hence, using the double prediction algorithm of the proposed MPC scheme, the optimal switching states, k_{selected} , and j_{selected} are chosen and the switching times, Δt_1 and Δt_2 are computed and thus are known at the next point of triggering. This in turn helps to achieve the objective of minimizing the switching frequency of single-phase IBC system at UPF along with the elimination of the flow of 100 Hz ripple power into the battery.

V. SIMULATION AND EXPERIMENTAL RESULTS

A. Simulation Results

The proposed control scheme has been simulated in MATLAB Simulink. The system parameters used for the simulation and experimental validation are given in Table II. The system has been tested for steady-state and transients in the system. In the present system, a resistive load is used as a dc load for both simulation studies and experimental verification. In the actual system, the battery shall replace the resistive load. Hence, in the present system, the capacitor ripple is a measure of harmonic power flow to the battery. A voltage controller is used to control the dc-link voltage. The voltage controller provides the magnitude of the reference grid current, and its angle is given by the single-phase PLL.

Fig. 3(b) shows the steady-state waveforms of the system. It can be seen that the grid current and the decoupling current are tracking the reference values. The grid voltage and the grid

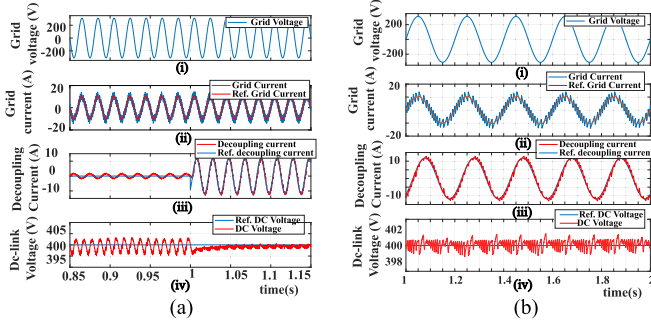


Fig. 3. (a) Transient results - Effect of decoupling. (b) Steady state waveforms (with decoupling). (i) Grid voltage (V), (ii) Actual and reference grid current (A), (iii) Actual and reference decoupling current (A), and (iv) Reference and actual dc-link voltage (V).

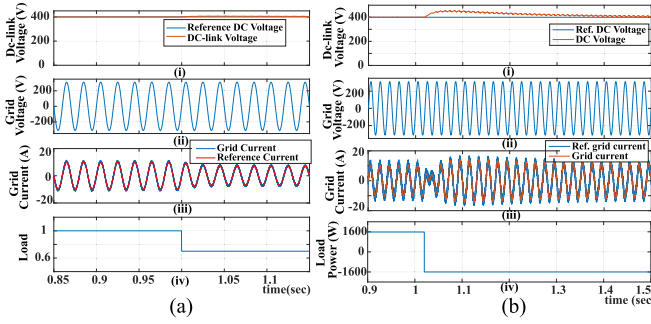


Fig. 4. Transient results - (a) Sudden change in load resistance: (i) Reference and actual DC-link voltage (V), (ii) Grid voltage (V), (iii) Actual and reference grid current (A), and (iv) Load. (b) Transition from G2V to V2G mode: (i) Reference and actual DC-link voltage (V), (ii) Grid voltage (V), (iii) Actual and reference grid current (A), and (iv) Load Power (W).

current are in phase with each other and thus, operating at unity power factor as shown in traces (i) and (ii) of Fig. 3(b). The torque produced in the motor is zero; hence, the motor does not rotate during the charging operation [18].

Fig. 3(a) shows the effect of active power decoupling on the dc-link voltage. The grid voltage waveform, the reference and the actual grid current waveforms, the current flowing in the c-phase winding, and the dc-link voltage waveform (zoomed version) are shown in Fig. 3(a). When the decoupling circuit is enabled at $t = 1$, it can be observed that there is a significant reduction in the 100 Hz ripple in the dc-link voltage as evident from trace-(iv) of Fig. 3(a), hence meeting the objective of 100 Hz voltage ripple reduction.

Fig. 4(a) shows the effect of load change on the grid current, decoupling current, and the dc-link voltage. It can be seen that as the load [trace-(iv)] decreases, the grid current [trace-(iii)] also reduces. The dc-link voltage remains constant with the change in the load, and the current controller tracks the changes in the reference grid current generated by the voltage controller. Fig. 4(b) shows the transition from grid-vehicle (G2V) mode to vehicle-grid (V2G) mode. It can be seen that when the transition happens, the phase of the grid current [trace-(iii)] reverses, and the actual current follows the reference current. The dc-link voltage [trace-(i)] is also maintained constant. Hence, it can

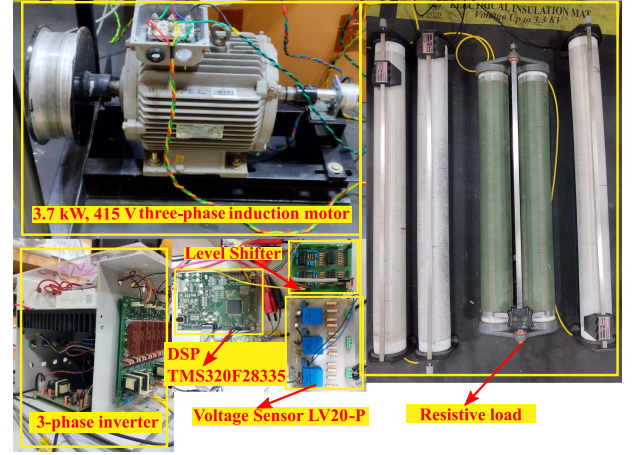


Fig. 5. Photograph of the experimental setup.

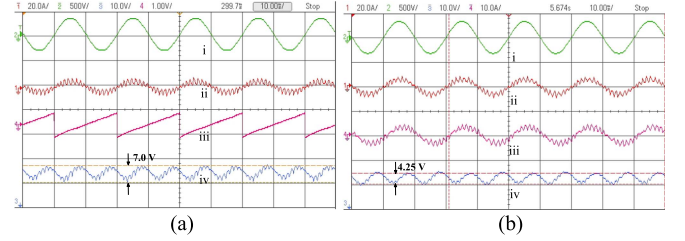


Fig. 6. (a) Steady-state results without decoupling. (b) Steady state results with active power decoupling. (i) Grid voltage (500 V/div), (ii) Grid current (20 A/div), (iii) Grid angle (1 V/div), and (iv) DC-link voltage (10 V/div).

be said that the proposed control scheme can be operated in vehicle-to-grid (V2G) mode as well.

B. Experimental Results

The proposed scheme is experimentally verified on 3.7 kW 415 V 3- ϕ induction motor with an open-end winding configuration. The semiconductor switches that were used to realize the inverter are 75-A, 1200-V IGBT half-bridge modules (SKM75GB12T4), and the gate driver used is M57962 L. A digital signal controller (TMS320F28335) with a clock frequency of 150 MHz is used to implement the proposed control scheme. Fig. 5 shows the photograph of the experimental setup for validating the proposed control scheme. As mentioned earlier, in the present system, a resistive load is used as a dc load for both simulation studies and experimental verification. In the actual system, the battery shall replace the resistive load. In the present system, the capacitor voltage ripple is a measure of harmonic power flow to the battery.

Fig. 6(a) shows the steady state waveforms at dc-link voltage of 180 V, without active-power decoupling. It can be seen that the grid current [trace (ii)] is in phase with the grid voltage [trace (i)], hence, depicting unity power factor operation. The trace (iii) shows the grid angle obtained by the using single-phase PLL. It can be seen from trace (iv) that the dc-link voltage ripple is 7 V. The same set of waveforms with active power decoupling enabled is shown in Fig. 6(b). It is evident from the figure that

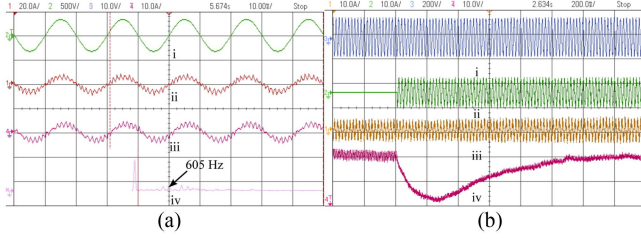


Fig. 7. (a) Steady-state results with decoupling. (i) Grid voltage (500 V/div), (ii) Grid current (20 A/div), (iii) Decoupling current (10 A/div), and (iv) FFT of grid current. (b) Effect of active power decoupling. (i) Grid voltage (200 V/div), (ii) Decoupling current (10 A/div), (iii) Grid current (10 A/div), and (iv) DC-link voltage (10 V/div).

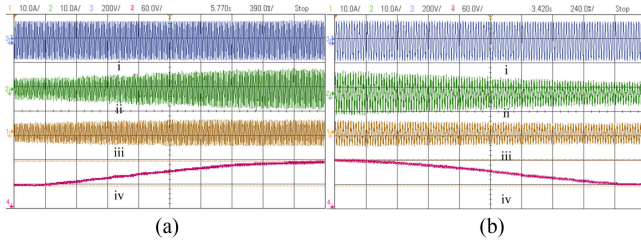


Fig. 8. Transient results- (a) Change in DC-link voltage from 180 to 240 V. (b) Change in DC-link voltage from 240 to 180 V. (i) Grid voltage (200 V/div), (ii) Grid current (10 A/div), (iii) Decoupling current (10 A/div), and (iv) DC-link voltage (60 V/div).

the ripple in the dc-link voltage is reduced to 4.25 V showing the effectiveness of the decoupling control using model predictive control scheme.

The frequency spectrum of grid current is shown in trace (iv) of Fig. 7(a). It can be seen that the lower-order harmonics are negligible compared to the fundamental component, and the major harmonic components are at 605 Hz. Fig. 7(b) shows the effect of the introduction of the decoupling circuit [represented by trace (ii), the current through the decoupling winding is zero till the active power decoupling mode is enabled]. With the introduction of the decoupling circuit, the reduction in the ripple of dc link voltage is evident from trace (iv) of the Fig. 7(b).

The transient operation of the system with the increase in the reference dc-link voltage from 180 to 240 V is shown Fig. 8(a). The dc-link voltage is tracking the reference voltage as can be seen from the figure. The waveforms corresponding to the decrease in dc-link voltage 240 to 180 V are shown in Fig. 8(b). The waveforms corresponding to transient operation during a step change in load from full load to half load condition at 180 V dc are given in Fig. 9(a). The controller is able to maintain the dc-link voltage as per the reference value. Similarly, the waveforms are obtained for a step change in load from half load to full load as shown in Fig. 9(b).

Typically in battery chargers, the capacitor connected in parallel to the battery would be around 300–500 μF . In the actual scenario, having a smaller or bigger capacitor across the battery will not make a difference, as the oscillating power (second harmonic) present on the dc bus will flow through the load (in this case, battery). The primary focus of our article is to mitigate

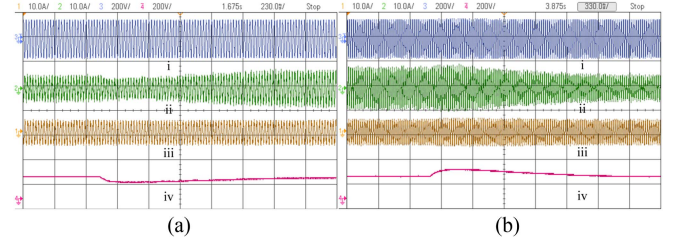


Fig. 9. Transient results. (a) Change in load from full load to half load. (b) Change in load from half load to full load. (i) Grid voltage (200 V/div), (ii) Grid current (10 A/div), (iii) Decoupling current (10 A/div), and (iv) DC-link voltage (200 V/div).

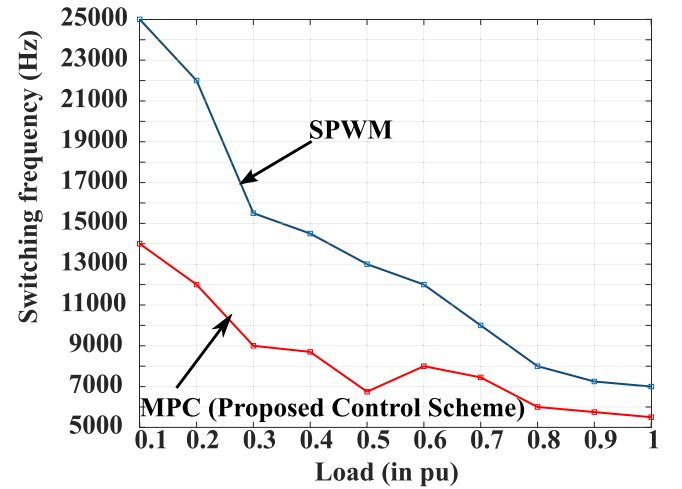


Fig. 10. Switching frequency variation with load for SPWM and MPC (proposed control scheme) for a fixed current ripple of 5%.

this oscillating power on the dc bus. To demonstrate this, we have opted for a resistive load in our lab experiments due to the unavailability of batteries.

The proposed model predictive control (MPC) scheme outperforms the sinusoidal pulse width modulation (SPWM) scheme. The current ripple for this analysis is quantified to be 5% of the grid current at any operating condition. So, as the grid current increases with increase in the load, the current ripple in the absolute sense will also be scaled appropriately. For a current ripple of 5% of the grid current, the switching frequency requirement in the case of SPWM was found for various load conditions. In a similar manner, the switching frequency for various load conditions in the proposed MPC scheme was also recorded for a current ripple of 5%. The switching frequency data for both cases is then plotted against the load. Fig. 10 illustrates the variation in switching frequency with load, aiming to uphold a consistent 5% ripple in current. It was observed that for the same current ripple (5%), a lower switching frequency operation is possible in the proposed MPC scheme compared to the conventional SPWM-based control scheme.

VI. CONCLUSION

In this article, a model predictive control scheme is proposed for the first time for the control of single-phase IBC. The MPC controller predicts and tracks the reference grid current and decoupling current waveforms. The current drawn by the grid is sinusoidal, and in phase with the grid voltage, hence, unity power factor operation is achieved. The issue of ripple power at the dc-link in the case of single-phase IBC is resolved by using active power decoupling. This is done by using the third winding of the motor for decoupling and controlling the current through it by using the MPC scheme. The ripple in the dc-link voltage is reduced to a great extent by using active power decoupling. In addition, zero torque production is also ensured during the charging mode. The MPC scheme presented in this article works on the double-prediction algorithm, which computes the optimum switching state and predicts the time instant at which the current error would go outside the boundary. The objective of the proposed MPC scheme to minimize the switching frequency is achieved, hence, minimizing the switching losses of the inverter and improving the efficiency of the system. A lower switching frequency operation is achieved by using the proposed MPC scheme compared to the SPWM-based control scheme for the same amount of current ripple. The vehicle-to-grid (V2G) mode of operation is also verified. The performance of the proposed scheme is validated by exhaustive experiments and simulation for steady-state and transient operation.

REFERENCES

- [1] "IEA, global EV outlook 2021: Accelerating ambitions despite the pandemic oecd, Paris, France," *IEEE Trans. Veh. Technol.*, 2021.
- [2] J. de Santiago et al., "Electrical motor drivelines in commercial all-electric vehicles: A review," *IEEE Trans. Veh. Technol.*, vol. 61, no. 2, pp. 475–484, Feb. 2012.
- [3] M. Y. Metwly, M. S. Abdel-Majeed, A. S. Abdel-Khalik, R. A. Hamdy, M. S. Hamad, and S. Ahmed, "A review of integrated on-board EV battery chargers: Advanced topologies, recent developments and optimal selection of FSCW slot/pole combination," *IEEE Access*, vol. 8, pp. 85216–85242, 2020.
- [4] T. Na, X. Yuan, J. Tang, and Q. Zhang, "A review of on-board integrated electric vehicles charger and a new single-phase integrated charger," *CPSS Trans. Power Electron. Appl.*, vol. 4, no. 4, pp. 288–298, 2019.
- [5] I. Subotic, N. Bodo, E. Levi, and M. Jones, "Onboard integrated battery charger for EVs using an asymmetrical nine-phase machine," *IEEE Trans. Ind. Electron.*, vol. 62, no. 5, pp. 3285–3295, May 2015.
- [6] I. Subotic, N. Bodo, and E. Levi, "Integration of six-phase EV drivetrains into battery charging process with direct grid connection," *IEEE Trans. Energy Convers.*, vol. 32, no. 3, pp. 1012–1022, Sep. 2017.
- [7] I. Subotic, N. Bodo, E. Levi, M. Jones, and V. Levi, "Isolated chargers for EVs incorporating six-phase machines," *IEEE Trans. Ind. Electron.*, vol. 63, no. 1, pp. 653–664, Jan. 2016.
- [8] I. Subotic, N. Bodo, and E. Levi, "An EV drive-train with integrated fast charging capability," *IEEE Trans. Power Electron.*, vol. 31, no. 2, pp. 1461–1471, Feb. 2016.
- [9] R. A. Taha et al., "Single-phase charging of six-phase integrated on-board battery charger using predictive current control," *IEEE Trans. Transport. Electrification*, to be published, doi: [10.1109/TTE.2023.3289089](https://doi.org/10.1109/TTE.2023.3289089).
- [10] A. S. Abdel-Khalik, S. Ahmed, and A. M. Massoud, "Performance evaluation of an on-board integrated battery charger system using a 12-slot/10-pole surface-mounted pm propulsion motor," in *Proc. 9th IEEE-GCC Conf. Exhib.*, 2017, pp. 1–9.
- [11] T. Wang, H. Wang, B. Xu, and J. Sui, "An integrated topology for battery charging and motor driving in electrical vehicle," in *Proc. IEEE Conf. Energy Internet Energy Syst. Integration*, 2017, pp. 1–5.
- [12] M. S. Diab, A. A. Elserougi, A. S. Abdel-Khalik, A. M. Massoud, and S. Ahmed, "A nine-switch-converter-based integrated motor drive and battery charger system for EVs using symmetrical six-phase machines," *IEEE Trans. Ind. Electron.*, vol. 63, no. 9, pp. 5326–5335, Sep. 2016.
- [13] C. Shi, Y. Tang, and A. Khaligh, "A three-phase integrated onboard charger for plug-in electric vehicles," *IEEE Trans. Power Electron.*, vol. 33, no. 6, pp. 4716–4725, Jun. 2018.
- [14] C. Viana and P. W. Lehn, "A drivetrain integrated DC fast charger with buck and boost functionality and simultaneous drive/charge capability," *IEEE Trans. Transport. Electrification*, vol. 5, no. 4, pp. 903–911, Dec. 2019.
- [15] V. F. Pires, J. Monteiro, A. Cordeiro, and J. F. Silva, "Integrated battery charger for electric vehicles based on a dual-inverter drive and a three-phase current rectifier," *Electronics*, vol. 8, no. 10, 2019, Art. no. 1199.
- [16] M. Kompella, P. Yadav, and R. S. Kaarthik, "A single phase integrated battery charger with active power decoupling for electric vehicles," in *Proc. IEEE Int. Conf. Power Electron. Drives Energy Syst.*, 2020, pp. 1–6.
- [17] H. R. Vangala and R. Kaarthik, "Control of a single phase integrated battery charger with active power decoupling for electric vehicles," in *Proc. IEEE Transp. Electrification Conf.*, 2021, pp. 1–6.
- [18] M. Kumar, C. S. Abhishek, and R. S. Kaarthik, "Current hysteresis control of a single phase integrated battery charger with active power decoupling," in *Proc. IEEE Int. Conf. Power Electron. Smart Grid Renewable Energy*, 2022, pp. 1–7.
- [19] D. Bhule and R. S. Kaarthik, "Model predictive control scheme for a single-phase integrated battery charger with active power decoupling for EV application," in *Proc. IEEE Int. Conf. Power Electron. Drives Energy Syst.*, 2022, pp. 1–6.
- [20] A. Habib et al., "Predictive current control of six-phase IM-based non-isolated integrated on-board battery charger under different winding configurations," *IEEE Trans. Power Electron.*, vol. 37, no. 7, pp. 8345–8358, Jul. 2022.
- [21] Q. Zhang, H. J. Raherimihaja, G. Xu, and X. Zhang, "Design and performance analysis of segmented three-phase IPMSM for EVs integrated battery charger," *IEEE Trans. Ind. Electron.*, vol. 68, no. 10, pp. 9114–9124, Oct. 2021.
- [22] S. Foti, A. Testa, G. Scelba, S. De Caro, and L. D. Tornello, "A V2G integrated battery charger based on an open end winding multilevel configuration," *IEEE Open J. Ind. Appl.*, vol. 1, pp. 216–226, 2020, doi: [10.1109/OJIA.2020.3035851](https://doi.org/10.1109/OJIA.2020.3035851).
- [23] H. J. Raherimihaja, Q. Zhang, T. Na, M. Shao, and J. Wang, "A three-phase integrated battery charger for EVs based on six-phase open-end winding machine," *IEEE Trans. Power Electron.*, vol. 35, no. 11, pp. 12122–12132, Nov. 2020.
- [24] M. Tong, M. Cheng, W. Hua, and S. Ding, "A single-phase on-board two-stage integrated battery charger for EVs based on a five-phase hybrid-excitation flux-switching machine," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 3793–3804, Apr. 2020.
- [25] D. Jiang, Y. Shuai, Z. Liu, B. Li, and M. Zhou, "Control strategy of series-end winding single-phase integrated on-board charger system," *IEEE Trans. Transport. Electrification*, to be published, doi: [10.1109/TTE.2023.3323525](https://doi.org/10.1109/TTE.2023.3323525).
- [26] G. K. N. Kumar and A. K. Verma, "A single-phase interleaved buck-boost PFC based on-board EV charger," *IEEE Trans. Ind. Appl.*, vol. 59, no. 6, pp. 7163–7174, Nov/Dec. 2023.
- [27] V. Shah and S. Payami, "Switched reluctance motor drive-train with fully integrated battery charger and instantaneous zero charging torque for electric transportation," *IEEE Trans. Transport. Electrification*, to be published, doi: [10.1109/TTE.2023.3308893](https://doi.org/10.1109/TTE.2023.3308893).
- [28] M. A. Khan, I. Husain, and Y. Sozer, "Integrated electric motor drive and power electronics for bidirectional power flow between electric vehicle and DC or AC grid," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2012, pp. 3403–3410.
- [29] M. Su, P. Pan, X. Long, Y. Sun, and J. Yang, "An active power-decoupling method for single-phase AC–DC converters," *IEEE Trans. Ind. Informat.*, vol. 10, no. 1, pp. 461–468, Feb. 2014.
- [30] H. Li, K. Zhang, H. Zhao, S. Fan, and J. Xiong, "Active power decoupling for high-power single-phase PWM rectifiers," *IEEE Trans. Power Electron.*, vol. 28, no. 3, pp. 1308–1319, Mar. 2013.
- [31] J. Holtz, "Advanced PWM and predictive control—An overview," *IEEE Trans. Ind. Electron.*, vol. 63, no. 6, pp. 3837–3844, Jun. 2016.
- [32] P. C. Krause, O. Wasynczuk, S. D. Sudhoff, and S. Pekarek, *Analysis of Electric Machinery and Drive Systems*, vol. 2. Hoboken, NJ, USA: Wiley, 2002.



Deeksha Bhule (Student Member, IEEE) received the B.E. degree in electrical engineering from Shri Shankaracharya Technical Campus, Bhilai, India, in 2017, and the M. Tech. degree in power system and control from the National Institute of Technology, Raipur, India, in 2020. She is currently working toward the Ph.D. degree with the Department of Avionics, Indian Institute of Space Science and Technology, Thiruvananthapuram, India.

Her research interests include the integrated battery chargers and control of multiphase motor drives.



R. Sudharshan Kaarthik (Senior Member, IEEE) received the B.Tech. degree in electrical engineering from the National Institute of Technology, Rourkela, India in 2010, the M.Tech. degree in electronics design and technology, and the Ph.D. degree in power electronics from the Indian Institute of Science, Bangalore, India, in 2012 and 2015, respectively.

Till 2017, he was a Postdoctoral Fellow with Concordia University, Montreal, QC, Canada. He is currently Associate Professor with the Department of Avionics, Indian Institute of Space Science and Technology, Thiruvananthapuram, India. His research interests include multilevel inverters, grid-connected systems, and electric motor drives.

Dr. Kaarthik is the recipient of the Indian National Academy of Engineering (INAE) Young Engineer Award in 2019, Indian Space Research Organization (ISRO) team award in 2018, Kerala State Young Scientist Award and Chief Minister's Gold Medal in 2020, Outstanding Researcher Award 2021 and Outstanding Professional Volunteer Award from IEEE Kerala Section for 2021 and 2022, respectively. He is an Associate Editor for IEEE TRANSACTIONS ON TRANSPORTATION ELECTRIFICATION, IEEE OPEN JOURNAL OF INDUSTRY APPLICATIONS, and *IEEE Transportation Electrification Community Newsletter*.